Neutrino Oscillations and the KamLAND Experiment

An example of quantum-mechanical dynamics leading to interference in a two-state system

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What are Neutrinos?

- Neutrinos are elementary particles with no charge and very small mass, much smaller than that of an electron. They interact only via the weak interaction and gravity, making them extremely difficult to detect as they pass through matter almost unimpeded.
- They occur in nature in three distinct flavours, electron neutrino (ν_e), muon neutrino (ν_μ) and tau neutrino (ν_τ).
- In quantum mechanics, a particle's state is represented as a vector (or "ket") in a Hilbert space. Thus, neutrinos are quantum states. We can describe two bases for neutrinos:
  1. Flavor states: $|\nu_e\rangle, |\nu_\mu\rangle, |\nu_\tau\rangle$
  2. Mass states: $|\nu_1\rangle, |\nu_2\rangle, |\nu_3\rangle$

Flavour vs Mass States of Neutrinos

- Flavour states are interaction eigenstates. They are the states that participate in weak interactions. However, the mass states are eigenstates of the free-particle Hamiltonian. Thus, when neutrinos propagate through space, their evolution depends on their mass.
- The flavour and mass eigenstates are not aligned, i.e. the basis that diagonalizes the interaction Hamiltonian is different from the basis that diagonalizes the free-particle (propagation) Hamiltonian. They are related by a unitary mixing matrix, called the PMNS (Pontecorvo–Maki–Nakagawa–Sakata matrix) matrix U, as follows:

$$|\nu_\alpha\rangle = \sum_{i=1}^{3} U_{\alpha i}|\nu_i\rangle$$

where $\alpha = e, \mu, \tau$ and $i = 1, 2, 3$.

Why Neutrino Oscillations happen?

- We will only be considering two flavours (ν_e and ν_μ) in this presentation to make it mathematically easier.
- Each flavour state is a combination of mass states, as follows:

$$|\nu_e\rangle = \cos\theta|\nu_1\rangle + \sin\theta|\nu_2\rangle$$
$$|\nu_\mu\rangle = -\sin\theta|\nu_1\rangle + \cos\theta|\nu_2\rangle$$

- In quantum mechanics, a stationary state evolves with time as:

$$|\psi(t)\rangle = e^{-iEt/\hbar}|\psi(0)\rangle$$

Each mass eigenstate $|\nu_i\rangle$ has its own energy E_i. So, if we start with an electron neutrino at time $t = 0$:

$$|\nu_e(0)\rangle = \cos\theta|\nu_1\rangle + \sin\theta|\nu_2\rangle$$

then after some time t, it becomes:

$$|\nu_e(t)\rangle = \cos\theta e^{-iE_1t/\hbar}|\nu_1\rangle + \sin\theta e^{-iE_2t/\hbar}|\nu_2\rangle$$

Why Neutrino Oscillations happen?

- We can use the inverse of our mixing relations to write $|\nu_1\rangle$ and $|\nu_2\rangle$ in terms of $|\nu_e\rangle$ and $|\nu_\mu\rangle$:

$$|\nu_1\rangle = \cos\theta|\nu_e\rangle - \sin\theta|\nu_\mu\rangle$$
$$|\nu_2\rangle = \sin\theta|\nu_e\rangle + \cos\theta|\nu_\mu\rangle$$

Now plug these into $|\psi(t)\rangle$:

$$|\psi(t)\rangle = \cos\theta\, e^{-iE_1 t/\hbar}(\cos\theta\,|\nu_e\rangle - \sin\theta\,|\nu_\mu\rangle) + \sin\theta\, e^{-iE_2 t/\hbar}(\sin\theta\,|\nu_e\rangle + \cos\theta\,|\nu_\mu\rangle)$$

Simplifying the coefficients of each flavor state:

$$|\psi(t)\rangle = \left[\cos^2\theta\, e^{-iE_1 t/\hbar} + \sin^2\theta\, e^{-iE_2 t/\hbar}\right]|\nu_e\rangle + \left[\cos\theta\,\sin\theta(e^{-iE_1 t/\hbar} - e^{-iE_2 t/\hbar})\right]|\nu_\mu\rangle$$

Deriving the Probability Functions

- The amplitude to find the initial electron neutrino as an electron neutrino is the coefficient of $|\nu_e\rangle$:

$$A(\nu_e \to \nu_e) = \cos^2\theta e^{-iE_1 t/\hbar} + \sin^2\theta e^{-iE_2 t/\hbar}$$

The probability is given by the modulus squared of the amplitude:

$$P(\nu_e \to \nu_e) = |A(\nu_e \to \nu_e)|^2$$

$$P(\nu_e \to \nu_e) = 1 - \sin^2 2\theta \sin^2\left(\frac{(E_2 - E_1)t}{2\hbar}\right)$$

- For a ultra-relativistic neutrino (speed $\approx c$), its energy is approximated as:

$$E = \sqrt{(pc)^2 + (mc^2)^2} \implies E_i \approx pc + \frac{m_i^2 c^3}{2p}, \qquad (1)$$

$$\text{thus, } E_2 - E_1 \approx \frac{\Delta m^2\, c^3}{2p}, \quad \text{where } \Delta m^2 = m_2^2 - m_1^2 \qquad (2)$$

Deriving the Probability Functions

- Now, since neutrinos travel close to the speed of light, $t \approx L/c$:

$$\frac{(E_2 - E_1)t}{2\hbar} \approx \frac{\Delta m^2 c^3 L}{4E\hbar}, \quad \text{using } E \approx pc \tag{3}$$

- We get the neutrino survival probability to be:

$$\boxed{P(\nu_e \to \nu_e) = 1 - \sin^2 2\theta \sin^2\left(\frac{\Delta m^2 L}{4E}\right)} \tag{4}$$

$$\text{taking natural units } \hbar = c = 1 \tag{5}$$

- The probability of detecting a muon neutrino instead is:

$$P(\nu_e \to \nu_\mu) = \sin^2 2\theta \sin^2\left(\frac{\Delta m^2 L}{4E}\right), \tag{6}$$

$$\text{since, } P(\nu_e \to \nu_e) + P(\nu_e \to \nu_\mu) = 1 \tag{7}$$

Mapping Neutrino Oscillations to the double-slit formalism

- Thus, the probability of finding the neutrino as a particular flavor depends on how the two mass components interfere - just like two waves interfering with a phase difference.
- This is a pure quantum interference pattern - oscillating between 0 and 1 as the phase difference builds up with distance.
- Neutrino oscillation is like a double-slit experiment, but in "mass space" instead of position space.
- The two mass eigenstates ($|\nu_1\rangle$ and $|\nu_2\rangle$) play the role of the two slits. The phase difference arising from L in the survival probability acts like the path length difference ΔL.
- The oscillating probability represents the quantum interference pattern, analogous to the spatial fringes observed on a detection screen for the double-slit experiment.

The KamLAND Experiment

- KamLAND stands for Kamioka Liquid-scintillator Anti-Neutrino Detector. It is a neutrino detector located deep underground in the Kamioka mine in Japan. KamLAND uses a large spherical tank filled with 1 kiloton of liquid scintillator to detect neutrino events.
- The site is surrounded by 53 Japanese commercial nuclear reactors. Nuclear reactors produce electron antineutrinos during the decay of the products formed during radioactive fission.
- When an electron anti-neutrino ($\bar{\nu}_e$) undergoes the inverse-beta decay reaction, i.e, interacts with a proton in the scintillator, it produces a positron and a neutron:

$$\bar{\nu}_e + p \rightarrow e^+ + n$$

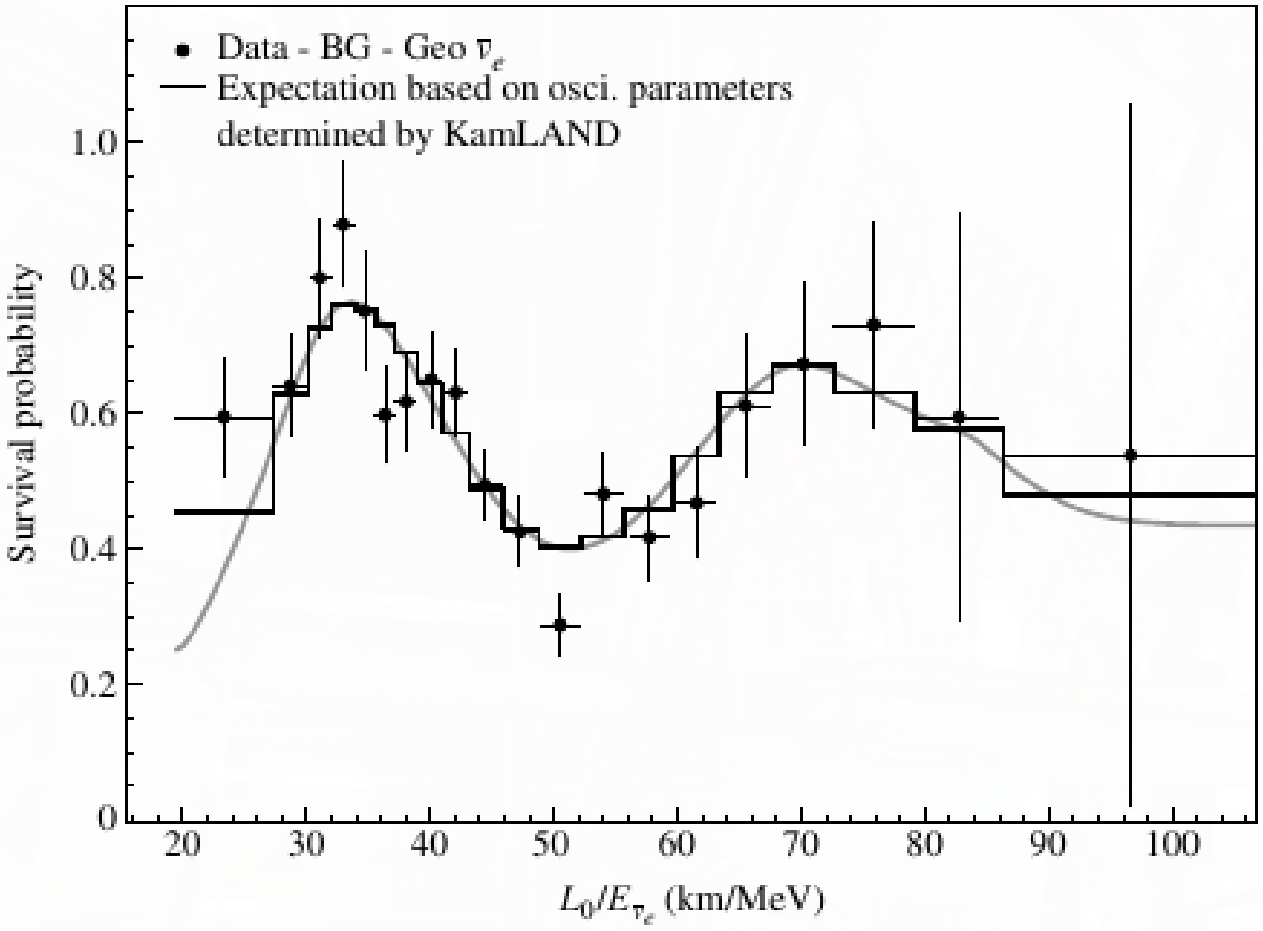
The KamLAND Experiment

- The positron creates an immediate light flash (prompt signal). The neutron is captured by another nucleus a few hundred microseconds later, releasing another light flash (delayed signal). This "delayed coincidence" signature makes $\bar{\nu}_e$ events easy to identify and distinguish from background noise.
- KamLAND reported that for 145 days of data, out of the 86.8 ± 5.6 events expected, only 54 were observed. And for 515 days of data, where 365.2 events were expected only 258 events were actually observed. These results antineutrino disappearance was established.
- Since neutrino's don't just vanish out of existence, the only way they can disappear is if they change into something the detector cannot see. KamLAND was designed only to detect electron neutrinos specifically, thus, it would not be able to detect the neutrinos if some of the neutrinos were to oscillate to muon or tau neutrinos.

The Oscillation Plot

- x-axis: L/E - distance divided by neutrino energy (in km/MeV). This ratio determines how much phase difference builds up between the mass states.
- y-axis: Survival probability - the probability that a $\bar{\nu}_e$ produced at a reactor is still a $\bar{\nu}_e$ when it reaches KamLAND.
- The data points (black dots) are the measured probabilities from KamLAND's observations, corrected for background signals (like from the Earth's natural radioactivity). The solid curve shows the theoretical prediction for neutrino oscillations using the best-fit oscillation parameters (Δm^2 and θ). The curve oscillates as L/E increases, just like a wave.

The Oscillation Plot



Why is the KamLAND Experiment Important?

- This experiment confirmed two crucial things: Neutrinos do oscillate (so flavor changes happen). Therefore, neutrinos must have mass, because oscillation requires non-zero mass differences ($\Delta m^2 \neq 0$).
- Neutrino oscillation is of great theoretical and experimental interest, as the precise properties of the process can shed light on several properties of the neutrino. In particular, it implies that the neutrino has a non-zero mass, which requires a modification to the Standard Model of particle physics.
- The experimental discovery of neutrino oscillation, and thus neutrino mass, by the Super-Kamiokande Observatory and the Sudbury Neutrino Observatories was recognized with the 2015 Nobel Prize for Physics.

Thank You!